

Tank Farm - Storage Tank & Pipe Rack

Oil & Gas Facility · 2026-09-30

OVERALL CONDITION	9	0	8	1
FAIR	Findings	P1 Immediate	P2 Planned	P3 Monitor

Source footage	oil_tank_farm_closeup.mp4
Footage length	19.4 s
Frames analysed	10 (one every 2.0 s)
Raw observations	16 (0 rejected as likely false alarms)
Detection model	Claude Opus 5.5 (frames reviewed in a Claude Code session, not via API)
Triage model	TypeSafe Jev (jev-1.13.0)
Processing time	6.9 s

How urgent is each priority

P1	Immediate (within 24-72 h)
P2	Planned (within 30 days)
P3	Scheduled / monitor (next maintenance cycle)

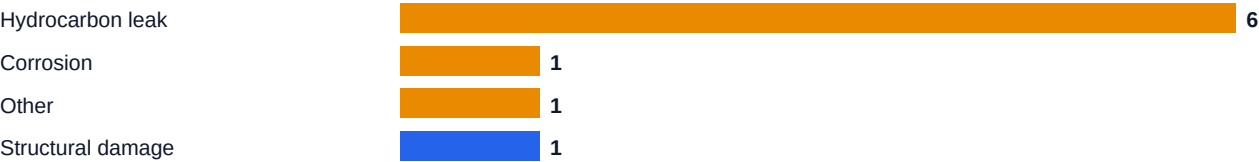
1. Executive summary

A 19-second close drone pass over one storage tank, its roof fittings and the adjacent ground-level pipe rack. The tank shell and pipework look sound and are well painted, but the ground tells a different story: dark stains cover the soil around the tank base (F-03) and follow the pipe rack for about 15 m (F-04), a pattern consistent with long-term drips from flanges, valves or drains. White deposits around the roof hatches (F-02) suggest vapour or product escaping through seals, and a small vertical vessel beside the tank is visibly rusting (F-06). A reddish, wet-looking drainage channel (F-01) and a dark band at the roof edge (F-08) need checking on site before any conclusion. None of the findings is an active emergency, but the staining should be tested promptly because it may mean a slow, ongoing release to soil.

Key risks

- F-03, F-04: Possible hydrocarbon contamination of soil around the tank and pipe rack from slow, repeated drips.
- F-02: Roof hatch or vent seals may be leaking vapour or product.
- F-01: Liquid in the drainage channel could be leaving the tank area (unconfirmed).
- F-06: Corrosion on the small vertical vessel will worsen without recoating.

Findings by category



2. Action plan

Immediate (24-72 hours)

- Take soil samples at the stained areas (F-03, F-04) and test for hydrocarbons.
- Walk the pipe rack and check every flange, valve and drain point above the stains for drips (F-04).
- Confirm the tank area's drain valves are closed and sample the drainage channel (F-01).

Next 30 days

- Inspect and replace roof hatch and vent seals; identify and clean the white deposits (F-02).
- UT-gauge and recoat the corroded vessel (F-06).
- Check the roof-to-shell seal (F-08) and reseal roof panel joints (F-05).
- Put the loose red equipment back in its proper place (F-07).

Next 90 days

- Remove any contaminated soil found by the sampling and record it for regulatory reporting.
- Seal the road crack (F-09) and re-fly the site to confirm fixes.

3. Findings register

All findings, most urgent first. Each is detailed with photos and repair steps in section 4.

ID	Time	Issue	Category	Severity	Priority	Action
F-01	00:18.0	Reddish-brown wet discolouration in drainage channel	hydrocarbon leak	Major (3.3)	P2	Clean
F-02	00:00.0	White crystalline deposits around roof hatches and stair landing	hydrocarbon leak	Major (3.0)	P2	Repair
F-03	00:02.0	Dark stains on soil around tank base	hydrocarbon leak	Major (2.9)	P2	Repair
F-04	00:06.0	Dark soil staining along pipe rack	hydrocarbon leak	Major (2.8)	P2	Repair
F-05	00:00.0	Dark streaking on dome roof panels	hydrocarbon leak	Moderate (2.0)	P2	Repair
F-06	00:08.0	Corroded small vertical vessel	corrosion	Moderate (2.0)	P2	Repair
F-07	00:14.0	Loose red equipment lying on the ground	other	Moderate (2.0)	P2	Repair
F-08	00:06.0	Dark band at roof-to-shell junction [verify on site]	hydrocarbon leak	Minor (1.2)	P2	Clean
F-09	00:16.0	Crack in road surface	structural damage	Minor (1.0)	P3	Repair

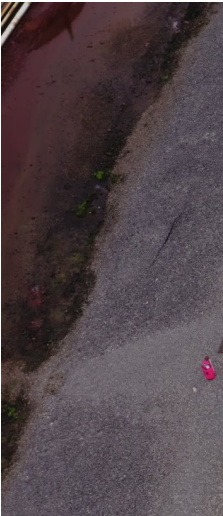
4. Detailed findings

F-01 Reddish-brown wet discolouration in drainage channel

P2 · Planned (within 30 days)



Full frame with approximate location



Zoomed detail

Timestamp	00:18.0 (seen in 1 frame(s), from 00:18.0)
Component	drainage channel beside the access road
Extent	Strip along the whole left side of the frame, ~15 m
Category	hydrocarbon leak (97% confidence)
Severity	Major, score 3.31 / 4
Safety / environmental hazard	41% probability
Likelihood it is a real defect	60%
First action	Clean: Remove dirt, debris, deposits, vegetation or spilled material; no component repair needed.
Crew	Operations + HSE + mechanical/pipefitter crew
Severity spread (Jev)	0 Cosmetic1 Minor2 Moderate3 Major4 Critical

What we saw

The drainage channel along the road has reddish-brown, wet-looking ground with weeds growing in it. It could be standing water with iron-rich sediment, or liquid draining from the tank area.

Visual evidence: Dark red-brown glossy surface along the channel, with green weeds along its edge.

How to fix it

For this finding: Sample the liquid or sediment in the channel; check where it drains from and that the tank area's drain valves are closed.

Standard procedure

- 1 Treat as a potential release: notify the site supervisor and HSE lead immediately.
- 2 Isolate the affected segment (close upstream/downstream block valves) and depressurise if safe.
- 3 Deploy absorbent booms/pads; contain spread to soil or water.
- 4 Locate the source (flange gasket, valve packing, pinhole, fitting) with a ground crew and gas detector.
- 5 Repair: re-torque or replace the gasket, repack the valve, or install an engineered clamp; replace the pipe spool for through-wall corrosion.
- 6 Excavate and dispose of contaminated soil; record volume released for regulatory reporting.

References: 40 CFR Part 112 (SPCC); API 570 Piping Inspection Code; API RP 1160 (pipelines)

F-02 White crystalline deposits around roof hatches and stair landing

P2 · Planned (within 30 days)



Full frame with approximate location

Zoomed detail

Timestamp	00:00.0 (seen in 1 frame(s), from 00:00.0)
Component	tank roof fittings and stair landing
Extent	Around 2-3 roof fittings and the landing, ~2 m2
Category	hydrocarbon leak (79% confidence)
Severity	Major, score 3.04 / 4
Safety / environmental hazard	40% probability
Likelihood it is a real defect	80%
First action	Repair: Fix the component in place (patch, recoat, re-torque, re-seal, re-route).
Crew	Operations + HSE + mechanical/pipefitter crew
Severity spread (Jev)	0 Cosmetic1 Minor2 Moderate3 Major4 Critical

What we saw

White powdery or crystalline deposits have built up around the roof hatch and vent fittings and on the stair landing below them. Deposits here often come from product vapour or liquid escaping through hatch or vent seals.
Visual evidence: Bright white granular patches concentrated around the circular roof fittings and spread across the landing grating.

How to fix it

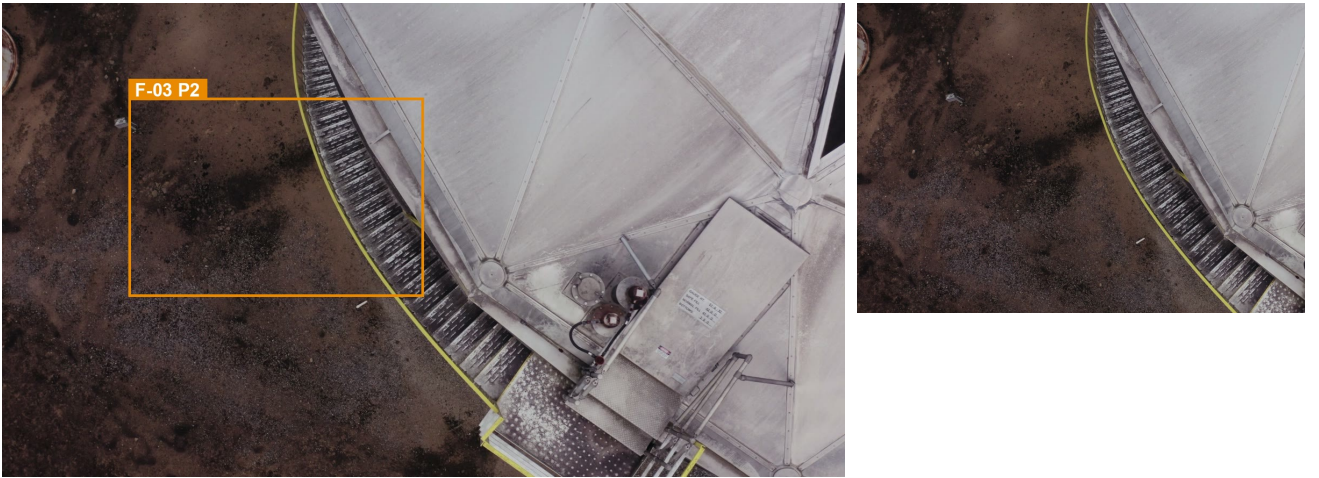
For this finding: Sample the deposit to identify it, then check the gaskets and seals on the roof hatches and vents; replace any that are worn and clean the deposits off.

Standard procedure

- 1 Treat as a potential release: notify the site supervisor and HSE lead immediately.
 - 2 Isolate the affected segment (close upstream/downstream block valves) and depressurise if safe.
 - 3 Deploy absorbent booms/pads; contain spread to soil or water.
 - 4 Locate the source (flange gasket, valve packing, pinhole, fitting) with a ground crew and gas detector.
 - 5 Repair: re-torque or replace the gasket, repack the valve, or install an engineered clamp; replace the pipe spool for through-wall corrosion.
 - 6 Excavate and dispose of contaminated soil; record volume released for regulatory reporting.
- References: 40 CFR Part 112 (SPCC); API 570 Piping Inspection Code; API RP 1160 (pipelines)

F-03 Dark stains on soil around tank base

P2 · Planned (within 30 days)



Full frame with approximate location

Zoomed detail

Timestamp	00:02.0 (seen in 2 frame(s), from 00:02.0)
Component	ground inside the tank area
Extent	Several patches over roughly 20-30 m2
Category	hydrocarbon leak (100% confidence)
Severity	Major, score 2.87 / 4
Safety / environmental hazard	53% probability
Likelihood it is a real defect	73%
First action	Repair: Fix the component in place (patch, recoat, re-torque, re-seal, re-route).
Crew	Operations + HSE + mechanical/pipefitter crew
Severity spread (Jev)	0 Cosmetic1 Minor2 Moderate3 Major4 Critical

What we saw

Irregular dark patches cover large parts of the soil around the tank base. They could be old product drips or spills soaked into the ground, or just damp soil.

Visual evidence: Dark, blotchy patches with soft edges on brown soil, darker than the surrounding ground and not matching any shadows.

How to fix it

For this finding: Take soil samples in the stained areas and test for hydrocarbons; if positive, find the source (buried line, flange or drain valve), repair it, and remove the contaminated soil.

Standard procedure

- 1 Treat as a potential release: notify the site supervisor and HSE lead immediately.
 - 2 Isolate the affected segment (close upstream/downstream block valves) and depressurise if safe.
 - 3 Deploy absorbent booms/pads; contain spread to soil or water.
 - 4 Locate the source (flange gasket, valve packing, pinhole, fitting) with a ground crew and gas detector.
 - 5 Repair: re-torque or replace the gasket, repack the valve, or install an engineered clamp; replace the pipe spool for through-wall corrosion.
 - 6 Excavate and dispose of contaminated soil; record volume released for regulatory reporting.
- References: 40 CFR Part 112 (SPCC); API 570 Piping Inspection Code; API RP 1160 (pipelines)

F-04 Dark soil staining along pipe rackP2 · Planned (within 30 days)



Full frame with approximate location

Zoomed detail

Timestamp	00:06.0 (seen in 4 frame(s), from 00:06.0)
Component	ground under and along the ground-level pipe rack
Extent	Along ~15 m of pipe rack
Category	hydrocarbon leak (100% confidence)
Severity	Major, score 2.77 / 4
Safety / environmental hazard	58% probability
Likelihood it is a real defect	84%
First action	Repair: Fix the component in place (patch, recoat, re-torque, re-seal, re-route).
Crew	Operations + HSE + mechanical/pipefitter crew
Severity spread (Jev)	0 Cosmetic1 Minor2 Moderate3 Major4 Critical

What we saw

Dark stains follow the line of the pipe rack, which sits low over bare soil. Staining concentrated under piping often means drips from flanges, valves or drains over time.

Visual evidence: Dark patches strongest directly beneath and beside the pipes, fading further away.

How to fix it

For this finding: Take soil samples in the stained areas and test for hydrocarbons; if positive, find the source (buried line, flange or drain valve), repair it, and remove the contaminated soil.

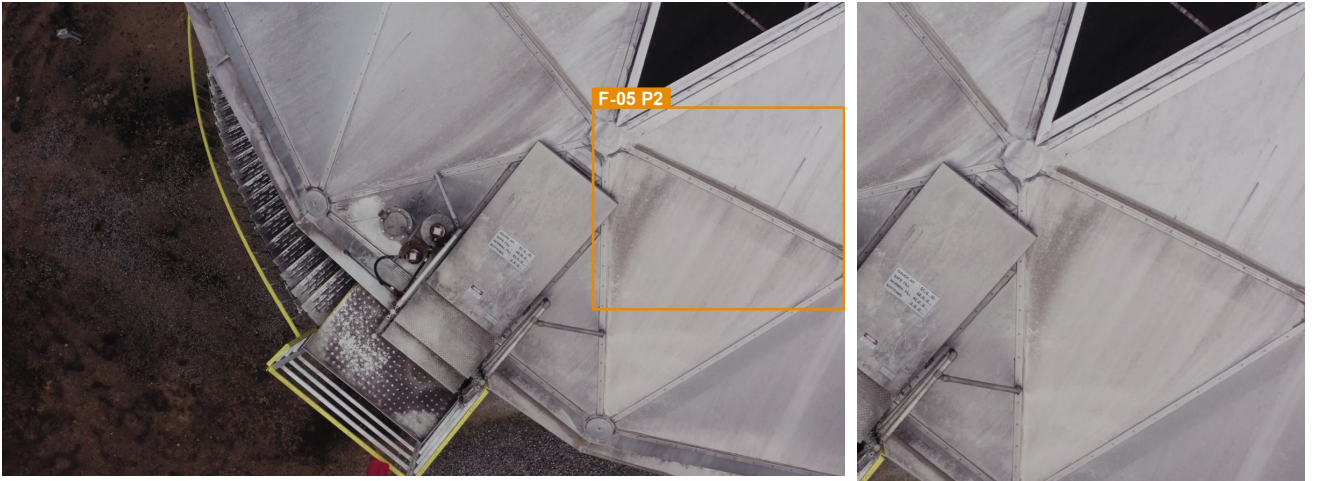
Standard procedure

- 1 Treat as a potential release: notify the site supervisor and HSE lead immediately.
- 2 Isolate the affected segment (close upstream/downstream block valves) and depressurise if safe.
- 3 Deploy absorbent booms/pads; contain spread to soil or water.
- 4 Locate the source (flange gasket, valve packing, pinhole, fitting) with a ground crew and gas detector.
- 5 Repair: re-torque or replace the gasket, repack the valve, or install an engineered clamp; replace the pipe spool for through-wall corrosion.
- 6 Excavate and dispose of contaminated soil; record volume released for regulatory reporting.

References: 40 CFR Part 112 (SPCC); API 570 Piping Inspection Code; API RP 1160 (pipelines)

F-05 Dark streaking on dome roof panels

P2 · Planned (within 30 days)



Full frame with approximate location

Zoomed detail

Timestamp	00:00.0 (seen in 1 frame(s), from 00:00.0)
Component	aluminium dome roof panels
Extent	Several panels on one side of the dome
Category	hydrocarbon leak (58% confidence)
Severity	Moderate, score 1.98 / 4
Safety / environmental hazard	17% probability
Likelihood it is a real defect	77%
First action	Repair: Fix the component in place (patch, recoat, re-torque, re-seal, re-route).
Crew	Operations + HSE + mechanical/pipefitter crew
Severity spread (Jev)	0 Cosmetic 1 Minor 2 Moderate 3 Major 4 Critical

What we saw

Several roof panels have dark vertical streaks running down from the panel joints.

Visual evidence: Grey-black streaks following the slope of the panels below the joint lines.

How to fix it

For this finding: Check the joints above the streaks for failed sealant; reseal where needed and clean the panels.

Standard procedure

- 1 Treat as a potential release: notify the site supervisor and HSE lead immediately.
- 2 Isolate the affected segment (close upstream/downstream block valves) and depressurise if safe.
- 3 Deploy absorbent booms/pads; contain spread to soil or water.
- 4 Locate the source (flange gasket, valve packing, pinhole, fitting) with a ground crew and gas detector.
- 5 Repair: re-torque or replace the gasket, repack the valve, or install an engineered clamp; replace the pipe spool for through-wall corrosion.
- 6 Excavate and dispose of contaminated soil; record volume released for regulatory reporting.

References: 40 CFR Part 112 (SPCC); API 570 Piping Inspection Code; API RP 1160 (pipelines)

F-06 Corroded small vertical vessel

P2 · Planned (within 30 days)



Full frame with approximate location

Zoomed detail

Timestamp	00:08.0 (seen in 3 frame(s), from 00:04.0)
Component	small vertical vessel / standpipe beside the tank
Extent	Rust over much of the visible shell
Category	corrosion (100% confidence)
Severity	Moderate, score 2.00 / 4
Safety / environmental hazard	17% probability
Likelihood it is a real defect	91%
First action	Repair: Fix the component in place (patch, recoat, re-torque, re-seal, re-route).
Crew	Integrity / NDT inspector + coatings contractor
Severity spread (Jev)	0 Cosmetic1 Minor2 Moderate3 Major4 Critical

What we saw

Closer view of the same vessel: heavy rust streaks on the shell and rust on the top.
Visual evidence: Dense orange-brown streaks and patches on the shell.

How to fix it

For this finding: UT-gauge the vessel wall at the rusted areas; if thickness is fine, wire-brush or blast and recoat. If there is pitting, have an engineer assess it before it goes back in service.

Standard procedure

- 1 Send a ground team to measure remaining wall thickness (UT gauging) at the corroded area.
- 2 Compare against minimum required thickness; calculate remaining life and corrosion rate.
- 3 If above minimum: blast-clean to bare metal and apply a suitable protective coating system.
- 4 If below minimum or pitting is deep: install a sleeve/clamp or replace the pipe section.
- 5 Check cathodic protection readings on buried or tank-bottom sections.

References: API 570; API 571 (damage mechanisms); AMPP/NACE SP0169 (cathodic protection)

F-07 Loose red equipment lying on the ground

P2 · Planned (within 30 days)



Full frame with approximate location

Zoomed detail

Timestamp	00:14.0 (seen in 1 frame(s), from 00:14.0)
Component	housekeeping near crossover stairs
Extent	1 item
Category	other (95% confidence)
Severity	Moderate, score 1.96 / 4
Safety / environmental hazard	29% probability
Likelihood it is a real defect	77%
First action	Repair: Fix the component in place (patch, recoat, re-torque, re-seal, re-route).
Crew	Site engineer
Severity spread (Jev)	0 Cosmetic 1 Minor 2 Moderate 3 Major 4 Critical

What we saw

A small red item, possibly a portable fire extinguisher or gas detector, is lying loose on the gravel next to the crossover stairs rather than mounted or stored.

Visual evidence: Small bright red object on its side on the gravel.

How to fix it

For this finding: Identify the item; if it is a fire extinguisher, put it back on its bracket and check its inspection tag.

Standard procedure

- 1
- Have a qualified engineer review the frame and decide on an inspection or repair.

F-08 Dark band at roof-to-shell junctionP2 · Planned (within 30 days)



Full frame with approximate location

Zoomed detail

Timestamp	00:06.0 (seen in 1 frame(s), from 00:06.0)
Component	tank roof edge / rim
Extent	Along the visible part of the rim
Category	hydrocarbon leak (70% confidence)
Severity	Minor, score 1.23 / 4
Safety / environmental hazard	19% probability
Likelihood it is a real defect	49% - VERIFY ON SITE before work
First action	Clean: Remove dirt, debris, deposits, vegetation or spilled material; no component repair needed.
Crew	Operations + HSE + mechanical/pipefitter crew
Severity spread (Jev)	0 Cosmetic 1 Minor 2 Moderate 3 Major 4 Critical

What we saw

A dark band runs along the edge where the roof meets the tank shell. It could be staining from the seal area or just dirt that has collected there.

Visual evidence: Continuous dark grey band following the curve of the roof edge.

How to fix it

For this finding: Inspect the roof-to-shell seal from the wind girder or roof; check for vapour or product at the seal and repair if needed.

Standard procedure

- 1 Treat as a potential release: notify the site supervisor and HSE lead immediately.
- 2 Isolate the affected segment (close upstream/downstream block valves) and depressurise if safe.
- 3 Deploy absorbent booms/pads; contain spread to soil or water.
- 4 Locate the source (flange gasket, valve packing, pinhole, fitting) with a ground crew and gas detector.
- 5 Repair: re-torque or replace the gasket, repack the valve, or install an engineered clamp; replace the pipe spool for through-wall corrosion.
- 6 Excavate and dispose of contaminated soil; record volume released for regulatory reporting.

References: 40 CFR Part 112 (SPCC); API 570 Piping Inspection Code; API RP 1160 (pipelines)

F-09 Crack in road surface

P3 · Scheduled / monitor (next maintenance cycle)



Full frame with approximate location

Zoomed detail

Timestamp	00:16.0 (seen in 2 frame(s), from 00:16.0)
Component	access road beside the tank area
Extent	~2 m long
Category	structural damage (68% confidence)
Severity	Minor, score 1.00 / 4
Safety / environmental hazard	9% probability
Likelihood it is a real defect	75%
First action	Repair: Fix the component in place (patch, recoat, re-torque, re-seal, re-route).
Crew	Structural/mechanical engineer + fabrication crew
Severity spread (Jev)	0 Cosmetic 1 Minor 2 Moderate 3 Major 4 Critical

What we saw

A thin crack runs across the access road surface.
Visual evidence: Dark hairline crack in the grey road surface.

How to fix it

For this finding: Seal the crack at the next road maintenance; monitor for widening.

Standard procedure

- 1 Restrict access below or near the damaged structure until assessed.
- 2 Have a structural or mechanical engineer assess the dent, crack or support failure.
- 3 Install temporary supports if a line is sagging.
- 4 Repair or replace the support, or cut out and replace the damaged pipe or tank section.

References: API 570; API 653 (above-ground storage tanks)

5. Methodology and limitations

How this report was produced

- Frames were sampled from the footage (one every 2.0 s, 10 in total).
- Detection: each frame was examined by a vision-language model (Claude Opus 5.5 (frames reviewed in a Claude Code session, not via API)) that listed every visible defect with its location, extent and visual evidence. It did not assign severity.
- Triage: each observation was scored by TypeSafe Jev (jev-1.13.0), a System One model that returns typed answers with probabilities: defect category, severity on a 5-level rubric, first action, the probability that the defect is genuine, and the probability of an immediate safety or environmental hazard.
- Rules applied by code: observations with less than 35% probability of being genuine were dropped; P1 if expected severity is 3.5 or higher or hazard probability is 60% or higher, or the recommended action is shutdown and severity is 2.5 or higher; P2 if severity is 1.75 or higher; otherwise P3. Findings with less than 60% probability of being genuine are capped at P2 and marked 'verify on site'.
- The same defect seen in neighbouring frames was merged into one finding.
- Repair procedures come from a reviewed playbook for this asset type; the per-finding fix is model-generated.

Severity scale

- 0: Cosmetic: no functional impact, note for records only.
- 1: Minor: early-stage wear; no loss of function; fix at next scheduled maintenance.
- 2: Moderate: measurable degradation or loss of output; fix within 30-90 days.
- 3: Major: active damage or significant loss of function; fix within 7-30 days.
- 4: Critical: active leak, fire, structural failure risk or danger to people or environment; act immediately.

Limitations

- Only what is visible from the air is assessed. Internal corrosion, wall thickness, electrical faults and gas leaks without a visible sign cannot be detected from RGB footage.
- Hotspot findings need radiometric thermal footage; on RGB footage they are not reliable.
- Locations are approximate boxes on the frame, not GPS coordinates.
- AI findings can be wrong. Every P1 and P2 finding should be confirmed on site before work starts.

Image credits

- Footage: "Drone Footage of a Circular Storage Tank", Pexels video #4324072 (Pexels License).